CSE2023 – HW2

1. $a = c \cdot k$, $b = d \cdot l$ $\quad ab = cdkl \rightarrow cd = \frac{ab}{kl}$ $\quad \frac{ab}{cd} = \frac{ab}{\frac{ab}{kl}} = kl$

2. a) $a = 23k + 40$ $\quad -22 \le 23k + 40 \le 0$ $\quad -62 \le 23k \le -40$ $\quad k = -2 \rightarrow a = -6$

b) $a = 29k + 15$ $\quad -14 \le 29k + 15 \le 14$ $\quad -29 \le 29k \le -1$ $\quad k = -1 \rightarrow a = -14$

c) $a \equiv 1 \pmod{21} \rightarrow a = 21k + 1$ $\quad 90 \le 21k + 1 \le 110$ $\quad k = 5 \rightarrow a = 106$

3. $n \pmod 4 \in \{0, 1, 2, 3\}$

$0^2 \pmod 4 \equiv 0$ $\quad 1^2 \pmod 4 \equiv 1$ $\quad 2^2 \pmod 4 \equiv 0$ $\quad 3^2 \pmod 4 \equiv 1$

4.

5. $5 = 4 \cdot 1 + 1$

$4 = 1 \cdot 4 + 0$

a) $\gcd(5, 4) = 1$

$200 = 121 \cdot 1 + 79$

$121 = 79 \cdot 1 + 42$

$79 = 42 \cdot 1 + 37$

$42 = 37 \cdot 1 + 5$

$37 = 5 \cdot 7 + 2$

$5 = 2 \cdot 2 + 1$

$2 = 1 \cdot 2 + 0$

b) $\gcd(200, 121) = 1$

$271 = 123 \cdot 2 + 25$

$123 = 25 \cdot 4 + 23$

$25 = 23 \cdot 1 + 2$

$23 = 2 \cdot 11 + 1$

$2 = 1 \cdot 2 + 0$

c) $\gcd(271, 123) = 1$

$14039 = 1529 \cdot 9 + 278$

$1529 = 278 \cdot 5 + 139$

$278 = 139 \cdot 2 + 0$

d) $\gcd(14039, 1529) = 139$

$14040 = 1530 \cdot 9 + 270$

$1530 = 270 \cdot 5 + 180$

$270 = 180 \cdot 1 + 90$

$180 = 90 \cdot 2 + 0$

$\gcd(14040, 1530) = 90$

6. $k = \{1, 2, 3, \dots b, \dots\}$ $\quad x = ak + b$

$k = b \rightarrow x = ab + b = b(a + 1)$

$b \mid x$ $\quad a + 1 \mid x$

7. a) $311 \equiv 7 \pmod{31}$ $918 \equiv 19 \pmod{31}$

$107 \equiv 7 \pmod{31}$ $100 \equiv 7 \pmod{31}$

$111 \equiv 18 \pmod{31}$ $310 \equiv 0 \pmod{31}$

b) $h(k,i) = h(k) + i \cdot \pmod{31}$, $0 < i < m-1$

8. $k1 \equiv 1524 \pmod{p}$ $k2 \equiv 578 \pmod{p}$ $h(k3,1) = (578 + 1927) \bmod p \equiv 2505$

$k4 \equiv 2376 \pmod{p}$ $k5 \equiv 3960 \pmod{p}$ $k6 \equiv 1526 \pmod{p}$

$k7 \equiv 2854 \pmod{p}$ $h(k8,1) = (1526 + 3401) \bmod p \equiv 4927$

$h(k9,1) = (3960 + 2140) \bmod p \equiv 1131$

$h(k10,2) = (3960 + 4280) \bmod p \equiv 3271$

9. (sum of first 10 digits) $\pmod 2 \equiv$ last digit

b has an error, last digit should be 1

10. a) 00 (00) 1111111 → 2 digits has an error

↓ but the check bit tells no error

00 (11) 0111111 (1)